

Dimensions and Units

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Outline

1. **Basic Concepts** – Dimensions vs. units, dimensional homogeneity
2. **Example 1: Pendulum Period** – Dimensional analysis step-by-step
3. **Example 2: Projectile Range** – Another dimensional analysis example
4. **Scaling Laws** – How dimensions predict scaling relationships

1 Basic Concepts

1.1 Dimensions vs. Units

Dimension: The *type* of quantity (length, mass, time, etc.)

Unit: A specific *scale* for measuring that dimension

Notation: $[x]$ denotes the dimension of quantity x

Fundamental dimensions:

$[L]$ = Length

$[M]$ = Mass

$[T]$ = Time

$[\Theta]$ = Temperature

Derived dimensions:

Velocity: $[v] = [LT^{-1}]$

Acceleration: $[a] = [LT^{-2}]$

Force: $[F] = [ma] = [MLT^{-2}]$

1.2 Dimensional Homogeneity

Golden Rule: You can only add, subtract, or equate quantities with the same dimensions.

Example 1: Valid

distance = speed $\times$ time

$$[L] = [LT^{-1}] \times [T] = [L] \quad \checkmark$$

Example 2: Invalid

$$\text{distance} = \text{speed} + \text{time}$$

$$[L] \neq [LT^{-1}] + [T] \quad \times$$

1.3 Transcendental Functions

Rule: Arguments to e^x , $\sin(x)$, $\ln(x)$, etc. must be **dimensionless**.

Example: In $N(t) = N_0 e^{kt}$

- kt must be dimensionless
- If $[t] = T$, then $[k] = T^{-1}$ (units: per time)

Example: In $\sin(\omega t)$

- ωt must be dimensionless (radians)
- If $[t] = T$, then $[\omega] = T^{-1}$ (radians per second)

2 Example 1: Pendulum Period (HW2 Problem 3)

2.1 Problem Statement

The period P of a simple pendulum depends on:

- Length L of the pendulum
- Gravitational acceleration g

Find the form of $P = f(L, g)$ using dimensional analysis.

2.2 Step-by-Step Solution

Step 1: List dimensions

$$[P] = T \quad (\text{period is time})$$

$$[L] = L \quad (\text{length})$$

$$[g] = LT^{-2} \quad (\text{acceleration})$$

Step 2: Assume power-law form

$$P = C \cdot L^a \cdot g^b$$

where C is a dimensionless constant, and a, b are exponents to determine.

Step 3: Write dimensional equation

$$[P] = [C] \cdot [L]^a \cdot [g]^b$$

Since C is dimensionless, $[C] = 1$ (no dimensions).

$$[P] = [L]^a \cdot [g]^b$$

Step 4: Substitute dimensions

$$T = L^a \cdot (LT^{-2})^b = L^a \cdot L^b T^{-2b} = L^{a+b} T^{-2b}$$

Step 5: Match dimensions on both sides For the equation $T = L^{a+b} T^{-2b}$ to hold, we need:

Length dimension:

$$0 = a + b \Rightarrow a = -b$$

Time dimension:

$$1 = -2b \Rightarrow b = -\frac{1}{2}$$

Step 6: Solve for a

$$a = -b = -\left(-\frac{1}{2}\right) = \frac{1}{2}$$

Step 7: Write final form

$$P = C \cdot L^{1/2} \cdot g^{-1/2} = C \cdot \frac{L^{1/2}}{g^{1/2}} = C \sqrt{\frac{L}{g}}$$

Step 8: Numerical example Given: $C = 2\pi$, $L = 0.75$ m, $g = 9.8$ m/s²

$$P = 2\pi \sqrt{\frac{0.75}{9.8}} = 2\pi \sqrt{0.0765} = 2\pi \times 0.277 \approx 1.74 \text{ seconds}$$

2.3 Excluding Mass (HW2 Problem 3b)

Question: Can the period depend on mass m ?

Step 1: Assume form with mass

$$P = C \cdot L^a \cdot g^b \cdot m^c$$

Step 2: Write dimensional equation

$$[P] = [L]^a \cdot [g]^b \cdot [m]^c$$

$$T = L^a \cdot (LT^{-2})^b \cdot M^c = L^{a+b} T^{-2b} M^c$$

Step 3: Match dimensions

$$\text{Length: } 0 = a + b$$

$$\text{Time: } 1 = -2b \Rightarrow b = -\frac{1}{2}, \quad a = \frac{1}{2}$$

$$\text{Mass: } 0 = c$$

Step 4: Conclusion Since $c = 0$, mass cannot appear in the formula. The period is independent of mass.

2.4 Scaling (HW2 Problem 3c)

Question: If length changes from L to $4L$, how does period change?

Step 1: Write ratio

$$\frac{P(4L)}{P(L)} = \frac{C\sqrt{4L/g}}{C\sqrt{L/g}} = \frac{\sqrt{4L}}{\sqrt{L}} = \frac{2\sqrt{L}}{\sqrt{L}} = 2$$

Step 2: Interpretation Doubling the length quadruples it, so period doubles: $P \propto \sqrt{L}$.

Step 3: Numerical example For $L = 3.0$ m (4 times 0.75 m):

$$P = 2\pi\sqrt{\frac{3.0}{9.8}} = 2\pi\sqrt{0.306} = 2\pi \times 0.553 \approx 3.48 \text{ seconds}$$

Check: $3.48/1.74 = 2.0$ (exactly double) ✓

2.5 Units Sanity Check (HW2 Problem 3d)

Question: Verify that using different units gives the same answer.

Step 1: Convert to centimeters

$$L = 0.75 \text{ m} = 75 \text{ cm}$$

$$g = 9.8 \text{ m/s}^2 = 980 \text{ cm/s}^2$$

Step 2: Compute period

$$P = 2\pi\sqrt{\frac{75}{980}} = 2\pi\sqrt{0.0765} = 2\pi \times 0.277 \approx 1.74 \text{ seconds}$$

Step 3: Verify Same answer as before! Units cancel correctly: $\sqrt{\frac{\text{cm}}{\text{cm/s}^2}} = \sqrt{\text{s}^2} = \text{s}$.

3 Example 2: Projectile Range

3.1 Problem Statement

A projectile is launched with initial speed v_0 at angle θ from horizontal. The range R (horizontal distance traveled) depends on:

- Initial speed v_0
- Gravitational acceleration g
- Launch angle θ (dimensionless)

Find the form of $R = f(v_0, g, \theta)$ using dimensional analysis.

3.2 Step-by-Step Solution

Step 1: List dimensions

$$\begin{aligned}[R] &= L \quad (\text{range is distance}) \\ [v_0] &= LT^{-1} \quad (\text{velocity}) \\ [g] &= LT^{-2} \quad (\text{acceleration}) \\ [\theta] &= 1 \quad (\text{angle is dimensionless})\end{aligned}$$

Step 2: Assume power-law form

$$R = C \cdot v_0^a \cdot g^b \cdot \theta^c$$

Since θ is dimensionless, it can appear in any form (e.g., $\sin \theta$, $\cos \theta$, etc.). For now, assume it appears as a dimensionless factor.

Step 3: Write dimensional equation

$$[R] = [v_0]^a \cdot [g]^b$$

(We ignore θ and C since they're dimensionless)

$$L = (LT^{-1})^a \cdot (LT^{-2})^b = L^a T^{-a} \cdot L^b T^{-2b} = L^{a+b} T^{-a-2b}$$

Step 4: Match dimensions

$$\begin{aligned}\text{Length: } 1 &= a + b \\ \text{Time: } 0 &= -a - 2b \quad \Rightarrow \quad a = -2b\end{aligned}$$

Step 5: Solve system Substitute $a = -2b$ into $1 = a + b$:

$$1 = -2b + b = -b \quad \Rightarrow \quad b = -1$$

$$a = -2(-1) = 2$$

Step 6: Write final form

$$R = C \cdot v_0^2 \cdot g^{-1} \cdot f(\theta) = C \frac{v_0^2}{g} \cdot f(\theta)$$

where $f(\theta)$ is some function of the angle (e.g., $\sin(2\theta)$).

Step 7: Physical interpretation

- Range increases as square of initial speed: $R \propto v_0^2$
- Range decreases as gravity increases: $R \propto 1/g$
- Range depends on launch angle through $f(\theta)$

Step 8: Known result The actual formula is $R = \frac{v_0^2 \sin(2\theta)}{g}$, which matches our dimensional analysis:

$$R = \frac{v_0^2}{g} \cdot \sin(2\theta)$$

3.3 Scaling Analysis

Question: If initial speed doubles, how does range change?

Step 1: Write ratio

$$\frac{R(2v_0)}{R(v_0)} = \frac{C(2v_0)^2/g \cdot f(\theta)}{Cv_0^2/g \cdot f(\theta)} = \frac{4v_0^2}{v_0^2} = 4$$

Step 2: Interpretation Doubling the initial speed quadruples the range: $R \propto v_0^2$.

4 Scaling Laws from Dimensions

4.1 General Principle

If a quantity Q has dimensions $[Q] = L^a M^b T^c$, then:

$$Q \propto (\text{length})^a \cdot (\text{mass})^b \cdot (\text{time})^c$$

4.2 Example: Pendulum Scaling

From $P = C\sqrt{L/g}$:

- If $L \rightarrow 4L$, then $P \rightarrow 2P$ (period doubles)
- If $g \rightarrow g/6$ (Moon), then $P \rightarrow \sqrt{6}P$ (period increases by $\sqrt{6}$)

4.3 Example: Surface Area to Volume Ratio

Problem: How does surface area to volume ratio scale with size?

Step 1: Dimensions

$$\begin{aligned} [\text{Surface area}] &= L^2 \\ [\text{Volume}] &= L^3 \\ \left[\frac{\text{Surface area}}{\text{Volume}} \right] &= \frac{L^2}{L^3} = L^{-1} \end{aligned}$$

Step 2: Scaling

$$\frac{\text{SA}}{\text{Vol}} \propto L^{-1} = \frac{1}{L}$$

Step 3: Interpretation Larger objects have relatively less surface area per unit volume. This affects:

- Heat loss (smaller animals lose heat faster)
- Nutrient absorption (smaller organisms have more surface relative to volume)

5 Summary

5.1 Key Principles

1. **Dimensional homogeneity:** Only quantities with same dimensions can be added/equated
2. **Transcendental functions:** Arguments must be dimensionless
3. **Power-law assumption:** For dimensional analysis, assume $Q = C \cdot x_1^{a_1} \cdot x_2^{a_2} \cdots$
4. **Match dimensions:** Set up equations for each fundamental dimension (L, M, T)
5. **Scaling:** Dimensions predict how quantities scale with size

5.2 Steps for Dimensional Analysis

1. List dimensions of all quantities
2. Assume power-law form: $Q = C \cdot x_1^{a_1} \cdot x_2^{a_2} \cdots$
3. Write dimensional equation: $[Q] = [x_1]^{a_1} \cdot [x_2]^{a_2} \cdots$
4. Expand and collect like dimensions

5. Match dimensions on both sides (set up system of equations)
6. Solve for exponents
7. Write final form
8. Verify with numerical example and units check

5.3 Key Points for HW2

- **Problem 3a:** Pendulum period $P = C\sqrt{L/g}$ (follow steps above)
- **Problem 3b:** Show mass cannot appear ($c = 0$ in $P = CL^a g^b m^c$)
- **Problem 3c:** Scaling ratio $\frac{P(4L)}{P(L)} = 2$
- **Problem 3d:** Units check: convert m to cm, verify same answer